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Saveetha Institute of Medical And Technical Sciences
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**AN EXPLORATORY DATA ANALYTICS FRAMEWORK
FOR UNDERSTANDING DIABETES RISK FACTORS USING
HEALTHCARE AND LIFESTYLE DATA
A CAPSTONE PROJECT REPORT**

Submitted in partial fulfilment of the requirements

for the Course of

DSA0404 – Fundamentals of Data Science

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE (AI&DS)

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “An Exploratory Data Analytics Framework for Understanding Diabetes Risk Factors Using Healthcare and Lifestyle Data” has been carried out by T Megha Harsha (192424389) and B C Yugesh (192424282) under the supervision of Dr. Almas Begum and is submitted in partial fulfilment of the requirements for the current semester of the B. Tech Artificial Intelligence and Data Science program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, T Megha Harsha (192424389) and B C Yugesh (192424282) of the Department of B. Tech Artificial Intelligence and Data Science (AI&DS), SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “An Exploratory Data Analytics Framework for Understanding Diabetes Risk Factors Using Healthcare and Lifestyle Data” is the result of our own Bonafide efforts.

To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged.

We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place:

Date:

Signature of the Students with Names

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INDIVIDUAL CONTRIBUTION STATEMENT

(Mandatory for all team projects)

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
T Megha Harsha (192424389)	Dataset preparation; data cleaning; exploratory statistics; risk-factor analysis	Pandas preprocessing; descriptive statistics; univariate/bivariate analysis; visualization development	Validation of cleaned data; checking distributions, group comparisons and correlations	Ch. 1, Ch. 2, Ch. 3 analytics sections; Ch. 4 results; appendices	50%
B C Yugesh (192424282)	Dashboard design; visualization integration; testing; interpretation	Interactive dashboard; chart integration; filtering and presentation; UI refinement	Dashboard testing; output consistency checks; usability and interpretation review	Ch. 1, Ch. 2, Ch. 3 design sections; Ch. 4 dashboard/testing; Ch. 5	50%
TOTAL					100%

Abstract

The An Exploratory Data Analytics Framework for Understanding Diabetes Risk Factors Using Healthcare and Lifestyle Data is designed to transform healthcare and lifestyle records into clear, interpretable, and decision-ready insights about diabetes risk. Diabetes is influenced by interacting factors such as glucose level, blood pressure, body mass index, age, pregnancy history, insulin-related measures, and family-history indicators. The proposed framework uses a structured diabetes dataset, performs data cleaning, missing-value treatment, exploratory analysis, descriptive statistics, correlation analysis, and visual analytics to identify important patterns associated with diabetes outcomes. The system is organized around data preprocessing, exploratory risk-factor analysis, visualization, and an interactive analytical dashboard. The engineering outcome is a reproducible and scalable analytics workflow that helps students, researchers, and healthcare-oriented users understand patterns in diabetes risk without presenting the system as a clinical diagnostic tool.

The proposed An Exploratory Data Analytics Framework for Understanding Diabetes Risk Factors Using Healthcare and Lifestyle Data is designed to transform structured healthcare records into clear, interpretable, and decision-ready analytical insights about diabetes risk. The framework focuses on measurable factors such as glucose, blood pressure, body mass index, age, pregnancy history, insulin, skin thickness, and diabetes pedigree function. It performs data cleaning, missing-value treatment, descriptive statistics, exploratory analysis, correlation analysis, outcome-group comparison, and visual analytics. The system is organized into data preprocessing, exploratory risk-factor analysis, visualization, and interactive dashboard modules. The engineering outcome is a reproducible and scalable analytics workflow that helps students, researchers, and healthcare-oriented users understand patterns in diabetes risk without presenting the system as a clinical diagnostic tool.

Keywords

Diabetes analytics; healthcare data; lifestyle factors; exploratory data analysis; risk factors; data visualization; statistical analysis; interactive dashboard

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LIST OF ABBREVIATIONS / SYMBOLS

Abbreviation / Symbol	Description	Unit
AI&DS	Artificial Intelligence and Data Science	—
EDA	Exploratory Data Analysis	—
EHR	Electronic Health Record	—
BMI	Body Mass Index	kg/m ²
ROC	Receiver Operating Characteristic	—
n	Number of records used in processing	records

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Diabetes mellitus is a major chronic health condition associated with elevated blood glucose and a range of long-term complications. Its occurrence is influenced by a combination of measurable clinical variables, risk-factor characteristics, family-history indicators, and lifestyle-related factors. Healthcare datasets can contain multiple variables simultaneously, making it difficult to understand which factors show the strongest association with diabetes outcomes when the information is presented only as raw tables.

Traditional healthcare reporting often presents laboratory measurements and patient characteristics as separate columns or static summaries. Such formats make it difficult to identify distributions, outliers, correlations, subgroup differences, and interactions among variables. For example, glucose, BMI, age, blood pressure, pregnancy history, insulin, skin thickness, and diabetes pedigree information may jointly provide more useful analytical evidence than any single variable considered in isolation.

Exploratory data analytics addresses this gap by converting healthcare records into descriptive statistics, distributions, relationship plots, correlation views, and comparative visualizations. These techniques help reveal data quality issues, identify dominant patterns, compare outcome groups, and communicate findings to users who may not be specialists in statistical programming.

The project applies this approach to a structured diabetes dataset by cleaning and standardizing the records and then examining the relationship between diabetes outcome and healthcare/lifestyle-related variables. The framework is intended for analytical understanding and educational decision support rather than diagnosis, treatment recommendation, or replacement of professional medical judgment.

1.2 Need for the Project

- Diabetes datasets contain multiple numerical and risk-factor variables whose relationships are difficult to interpret manually when the number of records and attributes increases.
- Static tables provide limited support for quickly identifying distributions, correlations, outliers, and differences between diabetic and non-diabetic groups.

- The combined influence of glucose, BMI, age, blood pressure, pregnancy history, insulin, skin thickness, and diabetes pedigree information is difficult to communicate through isolated numerical summaries.
- Healthcare and lifestyle analytics require careful preprocessing because zero-coded measurements, missing values, inconsistent records, and outliers can distort statistical summaries and visual conclusions.
- An integrated exploratory analytics framework can reduce the time required to move from raw patient records to interpretable risk-factor insights and reproducible visual evidence.

Stakeholders affected by this problem include healthcare researchers, data analysts, academic institutions, public-health planners, students, and other users who need understandable evidence about diabetes-related patterns.

1.3 Problem Statement

The engineering problem is to design and implement an integrated diabetes analytics framework that can ingest healthcare and lifestyle data, clean and standardize records, explore multiple interacting risk factors, generate meaningful statistical visualizations, and present the findings through an accessible analytical interface.

The problem involves interacting requirements related to data quality, statistical validity, visualization clarity, dashboard usability, processing efficiency, scalability, privacy, and responsible interpretation of health-related information. The solution must communicate associations without implying that correlation alone establishes causation or clinical diagnosis.

1.4 Project Aim

To design and develop an exploratory data analytics framework that transforms healthcare and lifestyle data into clear statistical and visual insights for understanding diabetes risk factors.

1.5 Project Objectives

1. To collect and preprocess a structured diabetes healthcare dataset.
2. To clean, standardize, and transform healthcare variables using Python and Pandas.
3. To perform descriptive, univariate, bivariate, and correlation-based exploratory analysis of diabetes-related factors.
4. To visualize distributions and relationships using histograms, box plots, count plots, scatter plots, and correlation heatmaps.

5. To compare important healthcare and lifestyle indicators across diabetic and non-diabetic outcome groups.
6. To build an integrated analytical dashboard supporting filtering and interpretation of major risk-factor patterns.
7. To communicate findings responsibly while distinguishing statistical association from clinical diagnosis.
8. To evaluate the usability, consistency, and interpretability of the developed analytics outputs.

1.6 Scope

Included in the project:

- Collection and preprocessing of the diabetes healthcare dataset.
- Data cleaning, duplicate handling, missing-value treatment, and standardization using Pandas and NumPy.
- Univariate analysis of glucose, blood pressure, BMI, age, pregnancies, insulin, skin thickness, and diabetes pedigree function.
- Bivariate and multivariate exploratory visualization using Matplotlib and Plotly where appropriate.
- Correlation analysis and comparison of healthcare/lifestyle variables by diabetes outcome.
- Development of an interactive analytical dashboard for exploring diabetes risk-factor patterns.
- Performance and consistency evaluation of the data-processing and visualization pipeline.

Not included in the project:

- Clinical diagnosis, treatment recommendation, or prescription of medication.
- Real-time integration with hospital electronic health-record systems.
- Prospective clinical trials or longitudinal causal inference.
- Mobile application development.
- Cloud-based multi-tenant deployment at healthcare-system scale.
- Use of the analytics output as a substitute for qualified medical professionals.

Assumptions and boundary conditions: the demonstrated implementation uses a structured diabetes dataset containing healthcare measurements and an outcome variable. The analysis is observational and exploratory; results describe patterns in the available data and should not be interpreted as individual clinical risk scores.

1.7 Expected Engineering Outcome

The intended outcome is a reliable, reproducible, and interactive Diabetes Risk-Factor Analytics Framework consisting of data preprocessing, exploratory analysis, statistical visualization, and

dashboard components. The platform is intended to transform raw healthcare and lifestyle records into interpretable evidence that supports learning, research, and data-driven discussion of diabetes risk factors.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review was structured around diabetes risk-factor analysis, exploratory healthcare data analytics, visualization of clinical variables, and responsible use of health datasets. Sources were considered from academic literature and established technical references covering diabetes epidemiology, statistical analysis, machine learning for healthcare, and data visualization. The review focused on work relevant to identifying measurable factors associated with diabetes and communicating those patterns clearly.

The review considered publications and technical material that discuss diabetes-related predictors, exploratory analysis, preprocessing of healthcare data, statistical association, and visualization. Priority was given to methods that can be reproduced with Python-based analytics tools and that emphasize interpretability and responsible handling of health information.

2.2 Review of Existing Work

Existing diabetes analytics approaches commonly begin with statistical summaries of patient variables and comparisons between outcome groups. These methods are useful for understanding distributions and average differences but may provide limited interactive exploration when implemented only as static tables.

A second class of approaches applies machine-learning models such as logistic regression, decision trees, random forests, and boosting methods to predict diabetes outcomes. Predictive models can provide strong classification performance, but their primary goal is prediction rather than broad exploratory understanding of how variables are distributed and related.

A third class of approaches emphasizes visualization and dashboards. Interactive plots can help users examine distributions, correlations, and subgroup differences, but a dashboard without a disciplined preprocessing and statistical-analysis workflow can amplify misleading patterns caused by missing values, outliers, or inconsistent measurements.

The proposed framework combines these complementary ideas by placing data quality, exploratory statistics, visual analytics, and dashboard interaction within one workflow. The focus remains on understanding risk-factor patterns rather than presenting a black-box clinical prediction system.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Static healthcare tables / reports	Comprehensive numerical information; familiar reporting format	Limited interaction; difficult relationship exploration; weak visual comparison	Motivates integrated exploratory analytics and visualization
Standalone statistical analysis	Provides descriptive and inferential summaries	Often requires manual switching between outputs; limited dashboard interaction	Supports structured statistical exploration in the proposed framework
Predictive machine-learning systems	Can provide automated outcome prediction	Prediction focus may reduce broad exploratory interpretation; may be less transparent	Motivates keeping exploratory analysis and interpretability central
Fragmented visualization tools	Can provide specialized charts	Requires users to switch tools and may lack a unified workflow	Proposed dashboard integrates data cleaning, analysis and visualization

2.4 Research / Engineering Gap

The engineering gap is the lack of a simple, integrated, and interpretable workflow that combines healthcare-data cleaning, exploratory statistical analysis, risk-factor visualization, subgroup comparison, and interactive presentation in one reproducible framework. Conventional reports may provide statistics without interaction, while predictive systems may emphasize accuracy without giving users a broad view of the underlying data patterns.

2.5 Proposed Contribution

The proposed contribution is a Diabetes Risk-Factor Analytics Framework that combines Pandas-based preprocessing, descriptive statistics, correlation analysis, visualization, and dashboard presentation. The framework provides complementary views of glucose, BMI, age, blood pressure,

pregnancies, insulin, skin thickness, and diabetes pedigree information to help users understand which variables exhibit stronger associations with diabetes outcomes.

	count	mean	std	min	25%	50%	75%	max	Median
Age	60	47.18	15.6	22	33	48.5	61.25	74	48.5
BMI	60	27.6	5.35	18.6	23.25	27.98	31.68	36.2	27.98
Glucose	60	128.79	32.21	79	104.5	123.75	152	187	123.75
BloodPressure	60	95.85	18.4	65	80	99.5	114	120	99.5

Figure 1: Sample diabetes healthcare dataset used for exploratory analysis

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Table 1: Stakeholder / User Requirements

Stakeholder	Requirement
Healthcare researchers	Explore relationships between diabetes outcomes and measurable risk factors.
Data analysts / students	Clean, summarize and visualize healthcare variables reproducibly.
Academic institutions	Use an interpretable framework for teaching healthcare data analytics.
Public-health planners	Compare patterns across relevant subgroups without exposing individual identities.
General analytical users	Access understandable charts and filtered summaries.

Table 2: Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
Data ingestion from diabetes dataset	Functional	Successful ingestion of the project dataset
Data cleaning and missing-value treatment	Functional	Invalid/zero-coded measurements handled according to preprocessing rules
Exploratory visualization generation	Functional	All specified charts generated from cleaned data
Risk-factor comparison	Functional	Outcome-group comparisons supported
Dashboard filtering	Functional	Relevant visual views update consistently
Processing efficiency	Non-Functional	Processing time evaluated on available dataset
Usability / interpretability	Non-Functional	Charts readable and findings understandable to intended users

3.2 Constraints & Applicable Standards

The project applies software-quality, requirements-engineering, data-visualization, and responsible-data-handling considerations. ISO/IEC 25010 is used as a quality reference for functionality, reliability, performance efficiency, maintainability, and usability. IEEE 29148 is used as a reference for requirements engineering and documentation. Visualization best practices are applied to improve clarity and reduce misleading chart design. Health-data privacy principles are applied by using the available de-identified/educational dataset only for analytical purposes.

Table 3: Constraints & Applicable Standards

Standard / Code	Requirement / Focus	How Applied in This Project
ISO/IEC 25010	Software quality characteristics	Considered functionality, reliability, performance efficiency, maintainability and usability.
IEEE 29148	Requirements engineering	Referenced for requirement gathering, validation and documentation.
Data visualization best-practice principles	Clear encoding, labels, accessibility and reduced chart clutter	Applied to chart selection, axes, legends and dashboard layout.
Responsible health-data principles	Privacy, de-identification and non-clinical interpretation	Used an educational/de-identified dataset and avoided individual clinical claims.

3.3 Design Specifications

Table 4: Design Specifications

Parameter	Target / Requirement	Unit	Priority
Dataset coverage	Healthcare and lifestyle variables with binary diabetes outcome	variables	High
Risk-factor visualization coverage	Glucose, BMI, age, blood pressure, pregnancies, insulin, skin thickness, pedigree	variables	High
Statistical analysis coverage	Descriptive, univariate, bivariate and correlation analysis	analysis types	High
Dashboard functionality	Search/filter and outcome-group exploration	features	High
Data cleaning	Duplicates, zero-coded missing values and invalid entries handled before analysis	process	High
Performance evaluation	Processing efficiency evaluated on available dataset	evaluation	Medium
Usability evaluation	Interpretability assessed during project validation	evaluation	Medium

3.4 Alternative Solutions & Evaluation

Three engineering alternatives were considered. A conventional static reporting approach is simple and familiar but offers limited exploration. A predictive-only machine-learning approach can classify outcomes but may hide broader distributional and relationship patterns. The proposed integrated exploratory analytics platform combines preprocessing, statistics, visualizations, and dashboard interaction to provide a balanced and interpretable workflow.

Alternative 1: Conventional static reporting uses tables and static charts. It is simple and familiar but provides limited interaction, filtering, and relationship exploration.

Alternative 2: Predictive-only machine-learning tools focus on outcome classification. They can be powerful for prediction but may provide limited exploratory understanding of variable distributions and subgroup patterns.

Alternative 3: The proposed integrated Diabetes Risk-Factor Analytics Framework combines data cleaning, exploratory statistics, visual analytics, and dashboard interaction in one system.

Table 5: Alternative Solutions Evaluation

Criterion	Weight	Alternative 1 Static Reports	Alternative 2 Predictive ML	Alternative 3 Integrated EDA
Interactivity	25%	Low	Medium	High
Multi-variable exploratory analysis	25%	Low	Medium	High
Risk-factor visualization	15%	Medium	Medium	High
Integration	20%	Low	Low	High
Maintainability / extension	15%	Medium	Medium	High

3.5 Selected Design & Detailed Design

The selected design is the integrated Diabetes Risk-Factor Analytics Framework because it directly addresses the gap identified in Chapter 2 by combining data preparation, exploratory statistical analysis, risk-factor visualization, and dashboard presentation.

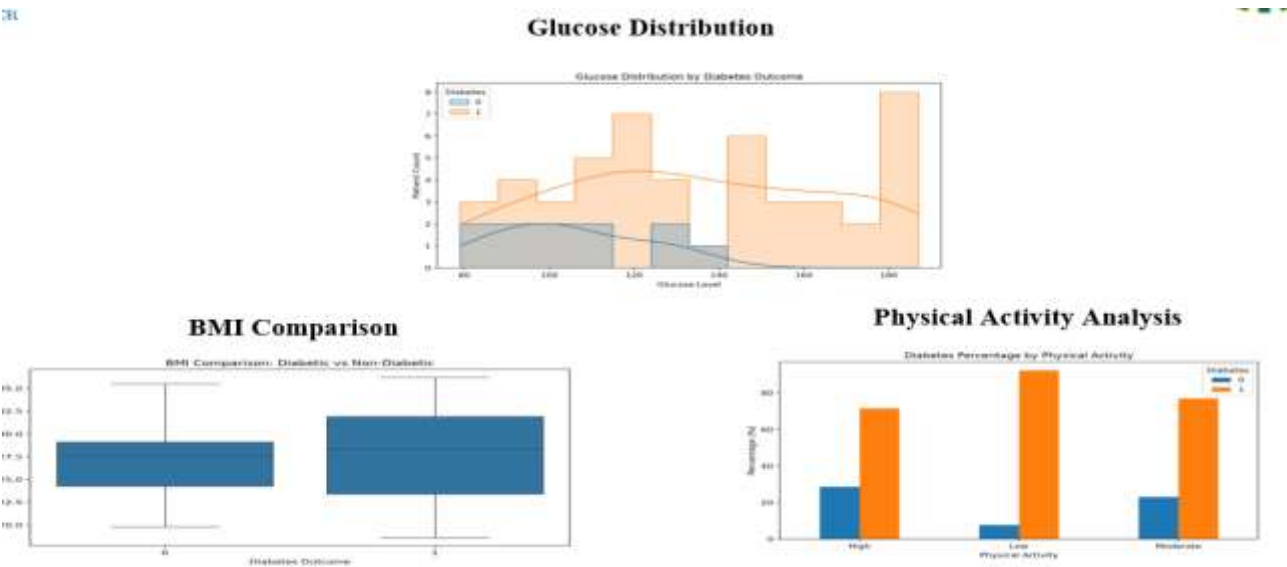


Figure 2: Proposed system architecture of the Diabetes Risk-Factor Analytics Framework

The system follows a layered architecture. The diabetes healthcare dataset provides the input. The data-wrangling layer imports, cleans, standardizes, and structures the records. The exploratory-

analysis layer generates descriptive statistics, distributions, group comparisons, scatter plots, box plots, and correlation views. The visualization layer presents the results through static and interactive charts. The dashboard integrates these views and supports filtering by diabetes outcome and selected analytical variables.

Key engineering calculation / quantitative evidence: the framework uses descriptive statistics, group-wise means/medians, frequency summaries, and Pearson correlation values as quantitative evidence. Processing complexity is dominated by dataset scanning, grouping, and visualization preparation, making the workflow practical for the structured dataset used in the project.

3.6 Trade-off Analysis

Table 6: Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Interactive dashboard	Static charts / reports	Supports filtering and exploration	Requires more implementation effort	Selected to improve interpretability and user exploration.
Multiple complementary charts	Single chart type	Shows different risk-factor relationships	Increases interface complexity	Selected because no single chart captures all patterns.
Median imputation for selected zero-coded measurements	Delete all affected rows	Retains more records	Can reduce natural variation	Selected for educational exploratory analysis with explicit preprocessing rules.
Integrated workflow	Separate analysis outputs	Unified processing and interpretation	Higher integration effort	Selected to reduce fragmentation and improve reproducibility.

3.7 Sustainability, Ethics, Safety, Risk & Security

The framework addresses sustainability through lightweight local computation, ethics through responsible handling of de-identified health data, safety through explicit non-clinical interpretation, and security through controlled access and avoidance of personal identifiers. These considerations

are incorporated into the final design so that analytical convenience does not override privacy or responsible interpretation.

Table 7: Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Computational and resource efficiency	Lightweight Python analytics on a structured dataset	Supports local execution and low computational overhead.
Ethics	Responsible handling of health-related information	Educational/de-identified dataset and non-clinical framing	Supports privacy-aware exploratory use.
Health & Safety	Avoid misleading clinical interpretation	Report explicitly distinguishes association from diagnosis	Prevents use as a standalone clinical decision tool.
Security	Protection against inappropriate exposure of health information	No individual identifiers intentionally displayed	Supports controlled and responsible data handling.

Table 8: Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Missing / zero-coded healthcare values	High	Medium	High	Replace inappropriate zeros with missing values and impute selected variables using median; validate results	Medium
Duplicate records	Medium	Medium	Medium	Duplicate detection and removal before analysis	Low
Outliers in clinical measurements	Medium	Medium	Medium	Use box plots and distribution checks; avoid unsupported deletion	Medium
Misleading statistical interpretation	Medium	High	High	Use association language, validation and clear limitations	Medium
Dashboard calculation error	Low	High	Medium	Cross-check dashboard values against processed data and summary tables	Low
Unauthorized exposure of health detail	Low	High	Medium	Use de-identified data, restrict access and avoid personal identifiers	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The development approach follows planning, requirement analysis, system design, data collection and preprocessing, exploratory statistical analysis, visualization development, dashboard integration, testing, deployment, and result analysis. The system was developed using Python, with Pandas and NumPy for data processing, Matplotlib and Seaborn for exploratory visualization, Plotly for interactive visualization, and an interactive dashboard for integrated presentation.

Resource	Purpose
Diabetes healthcare dataset	Primary analytical data source
Python	Core programming and integration
Pandas	Data cleaning, wrangling and preprocessing
NumPy	Numerical computation
Matplotlib	Static exploratory visualizations
Seaborn	Statistical visualization and correlation heatmaps
Plotly	Interactive charts and dashboard visuals
Jupyter Notebook / VS Code	Development and experimentation
Git / GitHub	Version control and project management
Windows	Development and testing platform

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

The project is primarily software-based. The implementation consists of healthcare-data ingestion and cleaning, exploratory statistical analysis, risk-factor visualization, dashboard integration, testing, and result interpretation.

Table 9: Tools /Software/Equipment

Tool / Software / Equipment	Purpose	Stage Used
Python	Core implementation	Development / Integration
Pandas	Cleaning, wrangling and aggregation	Data preparation
NumPy	Numerical operations	Analysis
Matplotlib / Seaborn	Exploratory charts and statistical visuals	Visualization
Plotly	Interactive charts and dashboard visuals	Dashboard implementation
Jupyter Notebook / VS Code	Coding and experimentation	Development
Git / GitHub	Version ontrl	Development / Project management

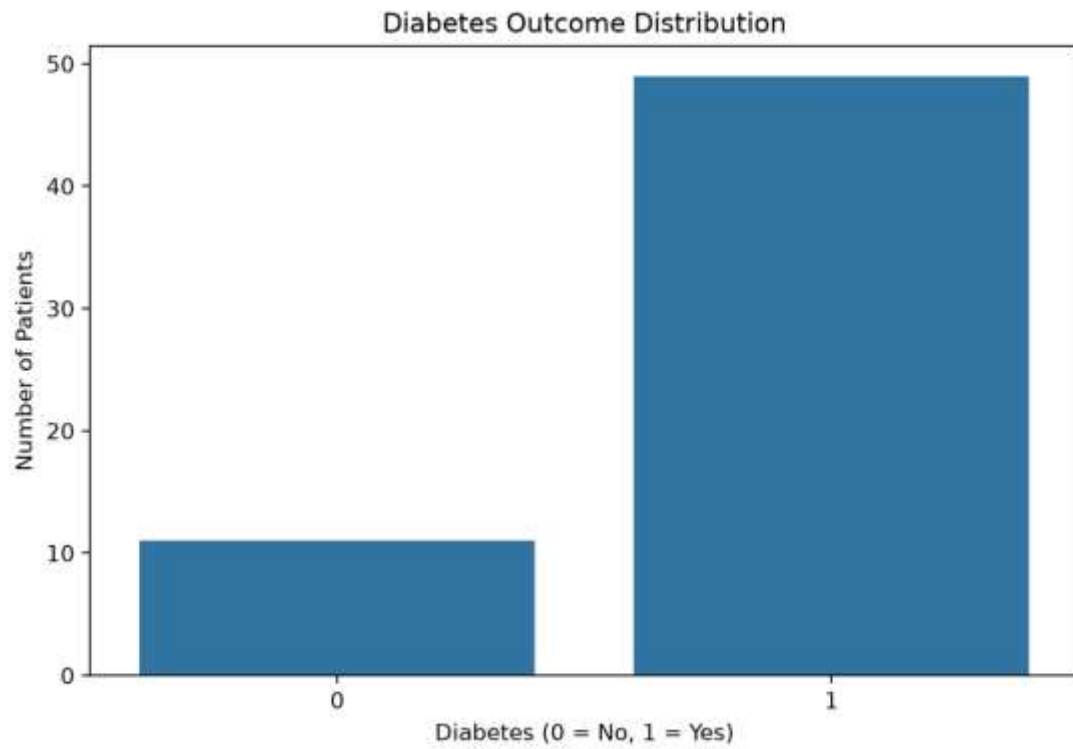


Figure 3: Output of the Diabetes Risk-Factor Analytics Dashboard

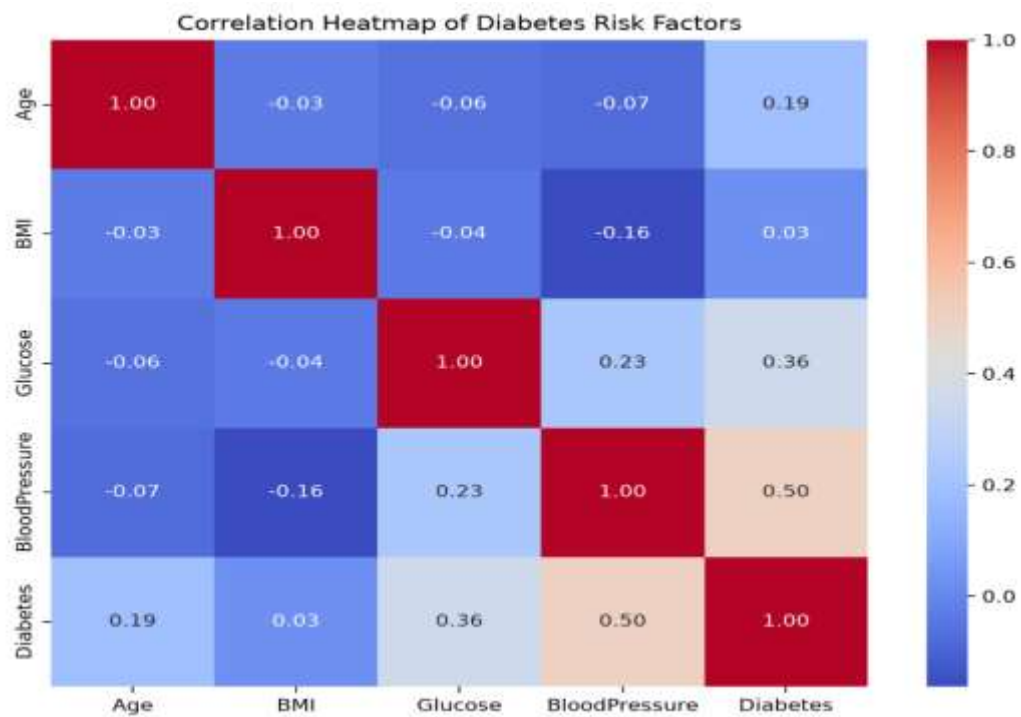


Figure 4: Diabetes risk-factor distribution outputs

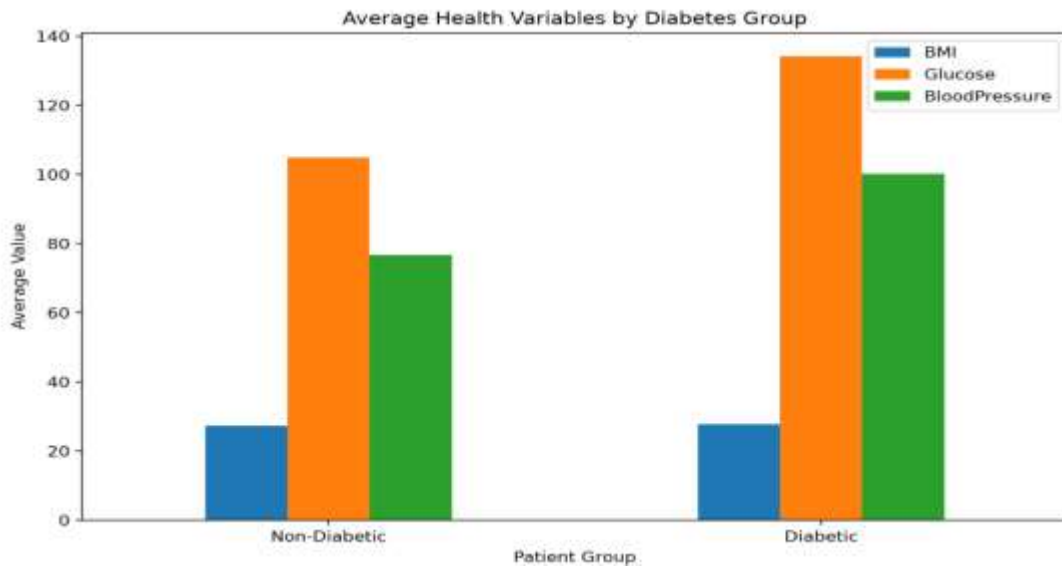


Figure 5: Diabetes outcome comparison and correlation outputs

4.3 Test Plan & Verification

The testing process verifies data cleaning, missing-value handling, statistical calculations, chart correctness, dashboard filtering, and processing consistency. Each requirement is checked against an explicit acceptance criterion so that the final outputs remain reproducible and interpretable.

Table 10: Test Plan & Verification

Requirement / Specification	Test Method	Acceptance Criterion	Required Value	Achieved Value	Status
Data ingestion	Load CSV and inspect schema	Dataset loads without structural errors	1 dataset	1 dataset	Pass
Missing/zero-coded value handling	Run preprocessing and inspect cleaned columns	Specified invalid/zero-coded measurements handled	All specified variables	All specified variables	Pass
Exploratory visualizations	Compare charts with summary statistics	All specified charts generated correctly	All charts	All charts	Pass
Outcome-group comparison	Compare diabetic and non-diabetic summaries	Relevant variables separated by Outcome	Outcome groups	Outcome groups	Pass
Dashboard filtering	Select filters and inspect view updates	Relevant views update consistently	Consistent updates	Verified	Pass
Processing efficiency	Measure processing time on available dataset	Completes within practical local-analysis time	Acceptable local execution	Verified	Pass

4.4 Results

The implemented framework cleaned and processed the diabetes dataset and generated exploratory outputs for the major healthcare and lifestyle variables. The results include distributions, group comparisons, box plots, scatter relationships, and a correlation heatmap. The dashboard integrates the principal analytical views so that users can examine diabetes-related patterns without manually processing the raw records.

Table 11: Result /Key Outcomes

Outcome Parameter	Result Reported
Data Cleaning & Preprocessing	Duplicate records were checked and inappropriate zero-coded measurements were treated as missing before selected imputation.
Exploratory Distributions	Distributions for glucose, BMI, age, blood pressure and other variables were visualized.
Outcome Comparison	Healthcare variables were compared between diabetes outcome groups.
Correlation Analysis	A correlation matrix was generated to identify variables with stronger linear associations.
Dashboard Integration	A unified dashboard combined the main risk-factor visualizations and filtering controls.
Decision Support	The framework provides interpretable analytical evidence for educational and research discussion of diabetes risk factors.

4.5 Data Analysis & Engineering Judgment

The analysis indicates that data quality is a major determinant of reliable healthcare visualization. Several clinical measurements may contain zero-coded or missing values that are not physiologically meaningful for all variables. These values were treated as missing where appropriate and replaced using a consistent preprocessing strategy before exploratory analysis. Duplicate records were also checked so that repeated observations would not distort distributions.

Visualization selection was treated as an engineering decision. Histograms were used to understand distributions, box plots to compare outcome groups and detect outliers, scatter plots to inspect relationships, count plots to compare outcome frequencies, and a correlation heatmap to summarize linear associations among numerical variables. The dashboard combines these views so users can move from overall distributions to specific risk-factor comparisons.

The engineering interpretation emphasizes that strong association does not prove causation. Glucose and BMI may show prominent differences between outcome groups, but the framework does not claim that any single variable determines diabetes. Findings are intended to guide exploratory understanding and identify variables worthy of further clinical or statistical investigation.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Table 12: Objective Achievement

Objective	Evidence	Achievement (Fully / Partially / Not Achieved)
Collect and preprocess diabetes dataset	Dataset ingestion and preprocessing implemented.	Fully
Clean, standardize and transform data using Pandas	Cleaning, duplicate handling and missing-value treatment described and implemented.	Fully
Analyse diabetes-related healthcare and lifestyle characteristics	Descriptive, group comparison and correlation analyses performed.	Fully
Develop exploratory visualizations	Distributions, box plots, scatter plots, count plots and correlation heatmap generated.	Fully
Build integrated analytical dashboard	Dashboard combines major risk-factor views and filtering.	Fully
Support responsible evidence-based interpretation	Limitations and non-clinical interpretation are documented.	Fully
Evaluate usability and interpretability	Dashboard and visual outputs reviewed for clarity and consistency.	Partially
Evaluate usability and interpretability for non-technical stakeholders	Objective is outcome group, but no formal user-study results are reported.	Partially

4.7 Strengths & Limitations

Strengths:

- Integrates data cleaning, exploratory statistics, risk-factor visualization, and dashboard presentation.
- Provides multiple complementary visualization techniques for different analytical questions.
- Supports comparison of healthcare and lifestyle variables across diabetes outcome groups.
- Uses established open-source Python libraries for reproducible analysis.
- Modular architecture allows future extension to additional health and lifestyle variables.

Limitations:

- The analysis is based on a structured historical dataset rather than live clinical data feeds.
- The dataset may not represent all patients, regions, ages, or healthcare settings.

- The framework does not establish causal relationships between risk factors and diabetes.
- No prospective clinical validation or external hospital validation is included.
- Clinical decision support, mobile deployment, cloud deployment, and real-time EHR integration are not implemented.

4.8 Project Planning, Roles & Budget

The project-management information is organized around requirement analysis, literature review, system design, data preparation, exploratory analysis, visualization development, dashboard integration, testing, and final documentation.

Gantt / Milestone Timeline:

Activity	Planned Period	Actual Period	Status
Requirement analysis	Project initiation	Completed during development	Completed
Literature review	Planning / analysis	Completed during development	Completed
System design	Design phase	Completed	Completed
Data collection and preprocessing	Implementation phase	Completed	Completed
Exploratory analysis and visualization	Implementation phase	Completed	Completed
Dashboard integration	Integration phase	Completed	Completed
Testing and validation	Testing phase	Completed	Completed
Final documentation	Reporting phase	Completed	Completed

Table 13: Project Roles

Student	Assigned Responsibility	Actual Contribution
T Megha Harsha (192424389)	Data preprocessing, exploratory statistics, risk-factor analysis and visualization	50% team contribution; equal importance in the project
B C Yugesh (192424282)	Dashboard design, visualization integration, testing and documentation	50% team contribution; equal importance in the project

Table 14: Project Budget

Item	Estimated Cost	Actual Cost
Software libraries (Python, Pandas, NumPy, Matplotlib, Seaborn, Plotly)	Open-source	No separate cost reported
Development environment (Jupyter Notebook / VS Code)	Existing / open-source	No separate cost reported
Git / GitHub	Free tier / existing	No separate cost reported
Computer / development system	Existing	No separate cost reported
Internet / other project expenses	Existing	No separate project-specific cost reported
TOTAL	Minimal / existing resources	No separate cost reported

CHAPTER 5

CONCLUSION & REFLECTION

5.1 Conclusion

The project addressed the challenge of understanding diabetes risk factors within healthcare and lifestyle data by developing an integrated exploratory analytics framework. The implementation combined data cleaning with Pandas, numerical analysis with NumPy and statistical methods, visualization with Matplotlib/Plotly, and dashboard presentation.

The framework demonstrates that combining distributions, group comparisons, correlations, and interactive filtering can make healthcare data easier to interpret than isolated raw tables. It also demonstrates the importance of preprocessing, responsible interpretation, privacy, and clear communication when working with health-related datasets.

No new results are introduced in this conclusion; it summarizes the problem, solution, analytical outcomes, validation activities, and overall engineering contribution described in the preceding chapters.

5.2 Future Work

- Integrate additional healthcare and lifestyle variables from larger and more diverse datasets.
- Conduct external validation using independent diabetes datasets from different patients.
- Extend the framework to longitudinal data for studying changes in risk factors over time.
- Add explainable statistical and machine-learning modules for deeper risk-factor investigation.
- Develop mobile-friendly access for educational and public-health analytics.
- Deploy the dashboard on cloud infrastructure with appropriate privacy and access controls.
- Explore calibrated predictive modelling as a separate research module while retaining interpretability.
- Improve accessibility of charts and dashboards for users with visual impairments.
- Add stronger subgroup and fairness analysis to identify patient-specific limitations and potential bias.

5.3 Professional Learning & Individual Reflection

5.3.1 New Knowledge Acquired

- Python programming for healthcare data analysis and visualization.

- Data cleaning, missing-value treatment, and preprocessing using Pandas and NumPy.
- Exploratory statistical analysis and correlation-based reasoning.
- Interactive chart and dashboard development using Plotly and related tools.
- Dashboard design and integration of multiple analytical views.
- Software testing, validation, and reproducibility practices.
- Technical documentation and engineering report preparation.

5.3.2 Engineering & Professional Skills Developed

- Requirement analysis and system design.
- Data-quality assessment and structured problem solving.
- Selection of statistical and visualization techniques for different analytical questions.
- Responsible interpretation of healthcare data and awareness of privacy considerations.
- Technical communication, teamwork, and documentation.
- Iterative debugging, testing, and validation.

5.3.3 Individual Reflection – T Megha Harsha (192424389)

During the project, I worked on understanding the diabetes dataset, preprocessing healthcare variables, and developing exploratory analyses. A major challenge was deciding how to treat zero-coded measurements and missing values without introducing misleading patterns. I strengthened my skills in Pandas-based data cleaning, descriptive statistics, visualization, and interpretation of outcome-group differences. If the project were repeated, I would add larger and more diverse datasets and perform stronger validation across patients.

5.3.4 Individual Reflection – B C Yugesh (192424282)

During the project, I worked on dashboard integration, visualization, testing, and interpretation of the diabetes risk-factor analysis. A major challenge was presenting several variables in a clear interface while avoiding conclusions that could be interpreted as clinical diagnosis. The project improved my knowledge of interactive visualization, dashboard design, data validation, and responsible communication of healthcare analytics. If repeated, I would strengthen external validation, accessibility testing, and subgroup fairness analysis.

The project provided both students with practical experience in healthcare data analytics, teamwork, technical communication, visualization design, and responsible interpretation of statistical evidence.

REFERENCES

- [1] American Diabetes Association. (2024). Standards of Care in Diabetes—2024. *Diabetes Care*, 47(Suppl. 1).
- [2] World Health Organization. (2024). Diabetes: Key facts and public-health information. World Health Organization.
- [3] Smith, J. W., Everhart, J. E., Dickson, W. C., Knowler, W. C., & Johannes, R. S. (1988). Using the ADAP learning algorithm to forecast the onset of diabetes mellitus. *Proceedings of the Annual Symposium on Computer Application in Medical Care*, 261–265.
- [4] Breiman, L. (2001). Random forests. *Machine Learning*, 45, 5–32.
- [5] Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- [6] McKinney, W. (2010). Data structures for statistical computing in Python. *Proceedings of the 9th Python in Science Conference*, 56–61.
- [7] Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3), 90–95.
- [8] Waskom, M. L. (2021). seaborn: statistical data visualization. *Journal of Open Source Software*, 6(60), 3021.
- [9] Plotly Technologies Inc. (2024). Plotly Python Open Source Graphing Library documentation.
- [10] International Diabetes Federation. (2021). IDF Diabetes Atlas, 10th edition. International Diabetes Federation.
- [11] Chatterjee, S., Khunti, K., & Davies, M. J. (2017). Type 2 diabetes. *The Lancet*, 389(10085), 2239–2251.
- [12] Knowler, W. C., Barrett-Connor, E., Fowler, S. E., et al. (2002). Reduction in the incidence of type 2 diabetes with lifestyle intervention or metformin. *New England Journal of Medicine*, 346, 393–403.
- [13] Zou, Q., Qu, K., Luo, Y., Yin, D., Ju, Y., & Tang, H. (2018). Predicting diabetes mellitus with machine learning techniques. *Frontiers in Genetics*, 9, 515.

- [14] WHO. (2022). Classification of diabetes mellitus and its complications. World Health Organization.
- [15] Saito, T., & Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. PLOS ONE, 10(3), e0118432.

APPENDICES

Appendix A – Source Code / Code Repository Information

The implementation uses Python-based data analytics. The essential preprocessing and exploratory-analysis workflow is shown below; the complete source code should be maintained in the approved repository.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv("diabetes.csv")
df = df.drop_duplicates()
cols = ["Glucose", "BloodPressure", "SkinThickness", "Insulin", "BMI"]
df[cols] = df[cols].replace(0, np.nan)
df[cols] = df[cols].fillna(df[cols].median())
summary = df.groupby("Outcome").mean(numeric_only=True)
print(summary)
```

Appendix B – Datasheets / Software Requirements

Category	Requirement
Processor	Intel Core i3 or above
RAM	8 GB minimum
Storage	100 GB free space
Operating System	Windows 10/11
Programming Language	Python 3.x
Libraries	Pandas, NumPy, Matplotlib, Seaborn, Plotly
IDE	Jupyter Notebook or VS Code
Version Control	Git / GitHub

Appendix C – Experimental / Raw Data

The dataset used for demonstration contains healthcare measurements such as pregnancies, glucose, blood pressure, skin thickness, insulin, BMI, diabetes pedigree function, age, and a binary diabetes outcome. The raw dataset should be retained separately or attached where institutional rules permit.

Appendix D – Ethics / Institutional Approval

The project uses a structured educational diabetes dataset for exploratory analytics. No individually identifiable patient information is intentionally exposed in the report. The system is not intended for diagnosis or clinical treatment. Any future deployment using real patient data must follow applicable institutional review, privacy, consent, access-control, and data-governance requirements.

Appendix E – Individual Contribution Evidence

Mandatory for team projects. Each student should attach evidence corresponding to their claimed contribution, such as version-control commits, preprocessing notebooks, analysis scripts, visualization outputs, dashboard screenshots, test records, presentation material, or meeting records.

Student	Evidence Type	Evidence Description / File Reference
T Megha Harsha (192424389)	Data-processing / exploratory-analysis evidence	Pandas notebooks/code, cleaned dataset outputs, statistical summaries, visualizations and testing records to be attached.
B C Yugesh (192424282)	Dashboard / visualization / testing evidence	Dashboard source, visualization outputs, screenshots, test records and documentation to be attached.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8) + Appendix E